Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

Experiment no:3

1. Aim: To deploy Multi-service application using Docker compoase

2. Theoretical Background

What is Docker Compose?

Docker Compose is an orchestration tool provided by Docker that allows developers to define and manage multi-container applications using a single YAML configuration file (`docker-compose.yml`). Instead of starting, networking, and linking individual containers manually using complex command-line arguments, Docker Compose allows you to configure all application services, network bridges, volume mounts, and environment variables in one structured file.

Key Concepts

- **Service:** An application component running in its own containerized environment (e.g., Node.js API, Nginx web server, PostgreSQL database).
- **Network:** An isolated internal bridge network that allows containers within the same stack to discover and communicate with each other using service names as domain names.
- **Volume:** A persistent storage mechanism managed by Docker that allows database files or cache data to persist even when containers are stopped, restarted, or deleted.
- **Environment Variables (.env):** A configuration file used to pass runtime parameters, credentials, and port configurations dynamically into containers without hardcoding them into source code.

3. Summary of Commands Executed

Below is the structured list of commands implemented throughout the lab, organized by their usage and lifecycle stage.

Category	Command Implemented	Description / Purpose
File Creation	`notepad docker-compose.yml`	Opens Windows Notepad to create and edit the YAML configuration file in the current working directory.

Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

File Creation	`notepad .env`	Creates the environment variable file to store dynamic settings like database passwords and port numbers.
Directory Setup	`mkdir backend` `mkdir frontend`	Creates structured folders to hold backend API source code, frontend assets, and custom Dockerfiles.
Troubleshooting	`ren backend\Dockerfile.txt Dockerfile`	Fixes Windows Notepad automatically appending .txt extensions to extensionless Dockerfiles.
Deployment	`docker compose up -d --build`	Forces Docker to rebuild custom container images, pull required registry images, and launch all services in detached mode.
Status Check	`docker compose ps`	Displays the current status, container names, health, and mapped host ports of all active services.
Testing	`curl http://localhost:3000`	Sends an HTTP request to test connectivity and output from the application endpoint in Command Prompt.
Log Inspection	`docker compose logs -f`	Streams aggregated live console logs from all containers for real-time output and error debugging.
Database Access	`docker compose exec postgres psql -U admin -d app_db`	Opens an interactive PostgreSQL database terminal inside the running postgres container.
Cache Access	`docker compose exec redis redis-cli`	Launches the Redis CLI inside the running cache container to inspect stored keys and visit counters.
Cleanup	`docker compose down -v`	Stops all running services, destroys container instances, deletes created bridges, and wipes associated data volumes.

Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

4. Step-by-Step Implementation Workflow

Step 1: Setting up Project Structure and Environment Variables

First, we configured the project workspace and stored configuration parameters inside a `.env` file to keep configurations separated from core application code.

```
FRONTEND_PORT=3000
BACKEND_PORT=5000
DB_PORT=5432
REDIS_PORT=6379
POSTGRES_USER=admin
POSTGRES_PASSWORD=secretpassword
POSTGRES_DB=app_db
```

Step 2: Configuring the Multi-Service YAML Stack

Next, we created `docker-compose.yml` to define our 3-tier setup combining frontend, backend, caching, and database services.

Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

```
services:   frontend:     build:
context: ./frontend
dockerfile: Dockerfile
container_name: app_frontend
ports:
-      "${FRONTEND_PORT}:80"     depends_on:
-      backend     networks:        - app-
network
   backend:
build:
      context: ./backend
dockerfile: Dockerfile
container_name: app_backend
ports:
-      "${BACKEND_PORT}:5000"
environment:
-      POSTGRES_USER=${POSTGRES_USER}
-      POSTGRES_PASSWORD=${POSTGRES_PASSWORD}
-      POSTGRES_DB=${POSTGRES_DB}
depends_on:
-      postgres       - redis     networks:
- app-network
```

Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

```
  postgres:
    image: postgres:15-alpine
container_name: app_postgres     ports:
-      "${DB_PORT}:5432"     environment:
-      POSTGRES_USER=${POSTGRES_USER}
-      POSTGRES_PASSWORD=${POSTGRES_PASSWORD}
-      POSTGRES_DB=${POSTGRES_DB}     volumes:
-      postgres_data:/var/lib/postgresql/data
networks:       - app-network
   redis:
    image: redis:alpine
container_name: app_redis     ports:
-      "${REDIS_PORT}:6379"     networks:
- app-network
 networks:
app-network:
    driver: bridge
 volumes:
postgres_data:
```

Step 3: Building and Deploying the Stack

We launched the stack using the build flag to ensure all local Dockerfiles were compiled into fresh container images.

```
C:\Users\15L\docker-compose-lab> docker compose up -d --build
```

Output:

Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

```
C:\Users\15L\docker-compose-lab> nano docker-compose.yml
'nano' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\15L\docker-compose-lab>notepad .env

C:\Users\15L\docker-compose-lab>mkdir web

C:\Users\15L\docker-compose-lab>notepad web\index.html

C:\Users\15L\docker-compose-lab>notepad web\Dockerfile

C:\Users\15L\docker-compose-lab>notepad docker-compose.yml

C:\Users\15L\docker-compose-lab>notepad docker-compose.yml
```

Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

```
C:\Users\15L\docker-compose-lab>notepad docker-compose.yml

C:\Users\15L\docker-compose-lab>docker compose up -d --build
#1 [internal] load local bake definitions
#1 reading from stdin 541B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 2B done
#2 DONE 0.0s
[+] up 0/1
 - Image docker-compose-lab-web Building                                                    0.5s
failed to solve: failed to read dockerfile: open Dockerfile: no such file or directory


C:\Users\15L\docker-compose-lab>dir web
 Volume in drive C is Windows
 Volume Serial Number is 2EA9-6C3A

 Directory of C:\Users\15L\docker-compose-lab\web

29-07-2026  13:31    <DIR>          .
29-07-2026  13:32    <DIR>          ..
29-07-2026  13:31                78 Dockerfile.txt
29-07-2026  13:31                50 index.html
               2 File(s)            128 bytes
               2 Dir(s)  736,351,133,696 bytes free

C:\Users\15L\docker-compose-lab>ren web\Dockerfile.txt Dockerfile

C:\Users\15L\docker-compose-lab>docker compose up -d --build
#1 [internal] load local bake definitions
#1 reading from stdin 541B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 115B done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/nginx:alpine
#3 DONE 2.5s

#4 [internal] load .dockerignore
```

```
C:\Users\15L\docker-compose-lab>notepad .env

C:\Users\15L\docker-compose-lab>docker compose up -d
[+] up 2/2
 ✔Container redis_cache    Running                                                          0.0s
 ✔Container custom_web_app Recreated                                                        0.1s

C:\Users\15L\docker-compose-lab>docker compose ps
NAME             IMAGE                    COMMAND                  SERVICE   CREATED          STATUS
        PORTS
custom_web_app   docker-compose-lab-web   "/docker-entrypoint.…"   web       15 seconds ago   Up 14 se
conds   0.0.0.0:8085->80/tcp, [::]:8085->80/tcp
redis_cache      redis:latest             "docker-entrypoint.s…"   redis     2 minutes ago    Up 2 min
utes    0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
```

Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

```
C:\Users\15L\docker-compose-lab>mkdir backend

C:\Users\15L\docker-compose-lab>notepad backend\package.json

C:\Users\15L\docker-compose-lab>notepad backend\server.js

C:\Users\15L\docker-compose-lab>notepad backend\Dockerfile

C:\Users\15L\docker-compose-lab>mkdir frontend

C:\Users\15L\docker-compose-lab>notepad frontend\index.html

C:\Users\15L\docker-compose-lab>notepad frontend\Dockerfile

C:\Users\15L\docker-compose-lab>notepad docker-compose.yml
```

```
 ✔Image docker-compose-lab-frontend       Built                                    14.9s
 ✔Image docker-compose-lab-backend        Built                                    14.9s
 ✔Volume docker-compose-lab_postgres_data Created                                  0.0s
 ✔Container app_postgres                  Created                                  0.1s
 ✔Container redis_cache                   Recreated                                0.4s
 ✔Container app_backend                   Created                                  0.1s
 ✔Container app_frontend                  Created                                  0.1s

C:\Users\15L\docker-compose-lab>docker compose ps
NAME            IMAGE                          COMMAND                  SERVICE    CREATED          ST
ATUS          PORTS
app_backend      docker-compose-lab-backend    "docker-entrypoint.s…"   backend    12 seconds ago   Up
 11 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
app_frontend     docker-compose-lab-frontend   "/docker-entrypoint.…"   frontend   12 seconds ago   Up
 11 seconds   0.0.0.0:3000->80/tcp, [::]:3000->80/tcp
app_postgres     postgres:15-alpine            "docker-entrypoint.s…"   postgres   13 seconds ago   Up
 11 seconds   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
app_redis        redis:alpine                  "docker-entrypoint.s…"   redis      13 seconds ago   Up
 11 seconds   0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
custom_web_app   docker-compose-lab-web        "/docker-entrypoint.…"   web        22 minutes ago   Up
 22 minutes   0.0.0.0:8085->80/tcp, [::]:8085->80/tcp
```

Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

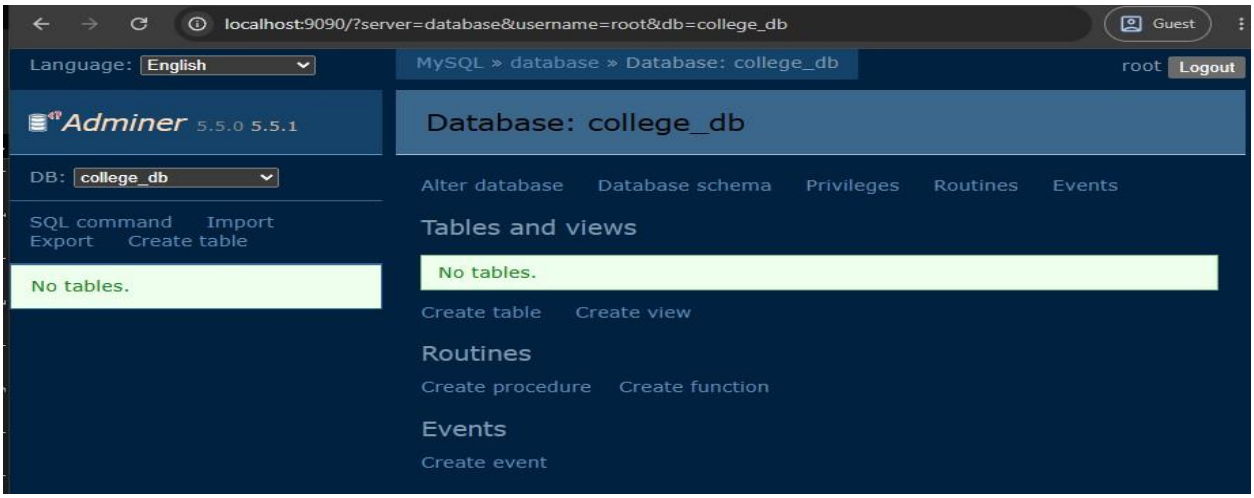
```
C:\Users\15L\docker-compose-lab>docker compose logs -f
app_backend    |
app_redis      | Starting Redis Server
app_frontend   | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform con
figuration
app_frontend   | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
app_frontend   | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
app_frontend   | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/defau
lt.conf
app_frontend   | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/def
ault.conf
app_frontend   | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
app_frontend   | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
app_frontend   | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
app_frontend   | /docker-entrypoint.sh: Configuration complete; ready for start up
app_frontend   | 2026/07/29 08:39:26 [notice] 1#1: using the "epoll" event method
app_postgres   | The files belonging to this database system will be owned by user "postgres".
app_backend    | > backend@1.0.0 start
app_backend    | > node server.js
app_backend    |
app_backend    | Backend listening on port 5000
app_redis      | 1:C 29 Jul 2026 08:39:26.588 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
app_redis      | 1:C 29 Jul 2026 08:39:26.588 * Redis version=8.8.1, bits=64, commit=00000000, modified
=1, pid=1, just started
app_redis      | 1:C 29 Jul 2026 08:39:26.588 * Configuration loaded
app_redis      | 1:M 29 Jul 2026 08:39:26.589 * monotonic clock: POSIX clock_gettime
app_redis      | 1:M 29 Jul 2026 08:39:26.590 * Running mode=standalone, port=6379.
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * <bf> RedisBloom version 8.8.2 (Git=unknown)
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * <bf> Registering configuration options: [
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * <bf>     { bf-error-rate       :      0.01 }
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * <bf>     { bf-initial-size     :       100 }
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * <bf>     { bf-expansion-factor :         2 }
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * <bf>     { cf-bucket-size      :         2 }
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * <bf>     { cf-initial-size     :      1024 }
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * <bf>     { cf-max-iterations   :        20 }
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * <bf>     { cf-expansion-factor :         1 }
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * <bf>     { cf-max-expansions   :        32 }
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * <bf> ]
app_redis      | 1:M 29 Jul 2026 08:39:26.591 * Module 'bf' loaded from /usr/local/lib/redis/modules//r
edisbloom.so
app_redis      | 1:M 29 Jul 2026 08:39:26.607 * <search> search-workers default: 16 (min of MAX_WORKER_
THREADS=16 and CPU cores)
app_redis      | 1:M 29 Jul 2026 08:39:26.607 * <search> Redis version found by RedisSearch : 8.8.1 - o
ss
app_redis      | 1:M 29 Jul 2026 08:39:26.607 * <search> RediSearch version 8.8.0 (Git=d2ae026)
app_redis      | 1:M 29 Jul 2026 08:39:26.607 * <search> Low level api version 1 initialized successful
ly
app_redis      | 1:M 29 Jul 2026 08:39:26.607 * <search> gc: ON, prefix min length: 2, min word length
```



Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

```
C:\Users\15L\docker-compose-lab>
C:\Users\15L\docker-compose-lab>docker compose logs -f backend
app_backend  |
app_backend  | > backend@1.0.0 start
app_backend  | > node server.js
app_backend  |
app_backend  | Backend listening on port 5000


C:\Users\15L\docker-compose-lab>docker compose logs postgres
app_postgres  | The files belonging to this database system will be owned by user "postgres".
app_postgres  | This user must also own the server process.
app_postgres  |
app_postgres  | The database cluster will be initialized with locale "en_US.utf8".
app_postgres  | The default database encoding has accordingly been set to "UTF8".
app_postgres  | The default text search configuration will be set to "english".
app_postgres  |
app_postgres  | Data page checksums are disabled.
app_postgres  |
app_postgres  | fixing permissions on existing directory /var/lib/postgresql/data ... ok
app_postgres  | creating subdirectories ... ok
app_postgres  | selecting dynamic shared memory implementation ... posix
app_postgres  | selecting default max_connections ... 100
app_postgres  | selecting default shared_buffers ... 128MB
app_postgres  | selecting default time zone ... UTC
app_postgres  | creating configuration files ... ok
app_postgres  | running bootstrap script ... ok
app_postgres  | sh: locale: not found
app_postgres  | 2026-07-29 08:39:26.966 UTC [35] WARNING:  no usable system locales were found
app_postgres  | performing post-bootstrap initialization ... ok
app_postgres  | syncing data to disk ... ok
app_postgres  |
app_postgres  |
app_postgres  | Success. You can now start the database server using:
```



Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

```
CONTAINER ID   NAME            CPU %     MEM USAGE / LIMIT     MEM %     NET I/O           BLOCK I/O
       PIDS
5f3a773690b8   app_frontend    0.00%     17.81MiB / 15.49GiB   0.11%     6.2kB / 2.88kB    1.67MB / 1
2.3kB   21
2d30636b5cca   app_backend     0.00%     43.14MiB / 15.49GiB   0.27%     11.2kB / 9.71kB   22.2MB / 4
.1kB    22
f8d0585d3183   app_postgres    0.00%     37.75MiB / 15.49GiB   0.24%     1.38kB / 126B     63.9MB / 5
3.4MB   6
fa95da0d4ea5   app_redis       0.24%     9.328MiB / 15.49GiB   0.06%     9.67kB / 8.25kB   58.7MB / 0
B       21
e3a83098ccab   custom_web_app  0.00%     18.97MiB / 15.49GiB   0.12%     4.4kB / 2.77kB    10.4MB / 1
2.3kB   21


C:\Users\15L\docker-compose-lab>docker compose exec postgres psql -U admin -d app_db
psql (15.18)
Type "help" for help.

app_db=#
app_db=#
app_db=# docker compose exec redis redis-cli
app_db-# \q

C:\Users\15L\docker-compose-lab>docker compose exec redis redis-cli
127.0.0.1:6379> KEYS*
(error) ERR unknown command 'KEYS*'
127.0.0.1:6379> *
(error) ERR unknown command '*'
127.0.0.1:6379> KEYS *
1) "visits"
127.0.0.1:6379> exit

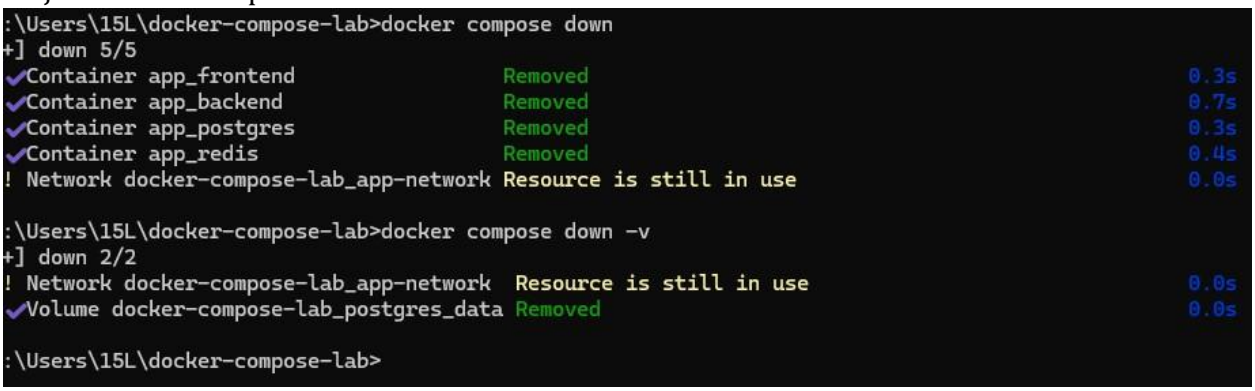
C:\Users\15L\docker-compose-lab>docker compose exec backend sh
/app #
/app # ^C

/app # exit

C:\Users\15L\docker-compose-lab>docker compose stop
[+] stop 4/4
 ✔Container app_frontend Stopped                                                    0.3s
 ✔Container app_backend  Stopped                                                    0.8s
 ✔Container app_redis    Stopped                                                    0.4s
 ✔Container app_postgres Stopped                                                    0.3s

C:\Users\15L\docker-compose-lab>docker compose start
[+] start 4/4
```

Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops



Name: Harsh Phale
Batch: T21
Roll no: 2401084
Subject: ASD&Devops

5. Conclusion

In this lab, we successfully built and deployed a multi-service application architecture using Docker Compose. By consolidating service definitions, network bridges, and volumes into a single `dockercompose.yml` file, we eliminated the need to run long, manual Docker commands for each service. Additionally, we addressed real-world deployment challenges such as port binding conflicts, file extension issues in Windows CMD, and image pulling over Docker Hub. The lab highlights how Docker Compose simplifies development workflows, improves local environment parity, and enables rapid deployment of complex microservices setups with minimal configuration overhead.